



Reg. No:

--	--	--	--	--	--	--	--	--	--	--

SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2025

COMMON TO B.E – CSE & M.TECH - CSE

20CS211 - ALGORITHMS AND COMPLEXITY-II

Answer ALL questions

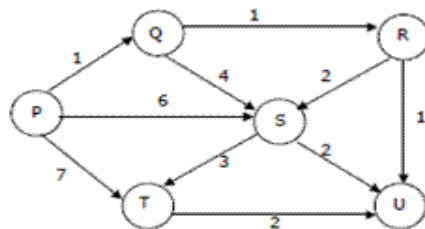
Duration: 3 Hours

Maximum: 100 Marks

PART - A

(7x 1 = 7 Marks)

- CO2** Choose the efficiency of the searching problem using transform-and-conquer algorithm design technique.
a) $\theta(n^2)$ b) $\theta(n)$ c) $\theta(n \log n)$ d) $\theta(\log n)$
- CO1** In a graph if $e=[u, v]$, Then u and v are called _____.
a) endpoints of e b) adjacent nodes c) neighbors d) edges
- CO1** The parent node of an i^{th} node in array will be at which index?
a) $i*2$ b) $i*2+1$ c) $i/2$ d) $i*2-1$
- CO2** Choose the requirements of Interpolation search.
a) unordered & linear b) linear & sorted
c) random & sequenced d) nonlinear & sorted
- CO2** Dijkstra's single source shortest-path algorithm on the following edge weighted directed graph with vertex P as the source. In what order do the nodes get included into the set of vertices for which the shortest path distances are finalized?



- a) P, Q, R, S, T, U b) P, Q, R, U, S, T c) P, Q, R, U, T, S d) P, Q, T, R, U, S

6. **CO1** Identify the name of a node in a state space tree, if it corresponds to a partially constructed solution that may still lead to a complete solution.
 a) Promising b) Non promising c) feasible d) optimal
7. **CO3** Assuming $P \neq NP$, which of the following is true ?
 a) NP-complete = NP b) NP-complete $\cap P = \Phi$
 c) NP-hard = NP d) $P = NP$ -complete

Fill in the blanks:

(3x 1 = 3 Marks)

8. **CO1** A complete binary tree of height h has a maximum of _____ Nodes.
9. **CO2** _____ is the Time complexity of Fake coin identification.
10. **CO1** _____ Tree is used for encoding and decoding.

PART - B

(5x 3 = 15 Marks)

11. **CO2** Compute 18×85 using Russian Peasant method.
12. **CO2** Solve the polynomial $P(x) = 5x^4 + 3x^3 - 2x^2 + 6x - 7$ at $(x = 5)$ using Horner's Rule.
13. **CO1** Construct 2-3 tree for the list 1,2,3,4,5,6.
14. **CO3** Describe the use of Reduction in solving NP complete Problems?.
15. **CO3** Indicate how many 3 vertex cliques are formed in a 4 vertex completely connected graph?

PART - C

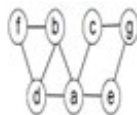
(5 x 15 = 75 Marks)

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

16. **CO1** Consider the graph, (15)



- Write down the adjacency matrix and adjacency lists specifying this graph. (Assume that the matrix rows and columns and vertices in the adjacency lists follow in the alphabetical order of the vertex labels.)

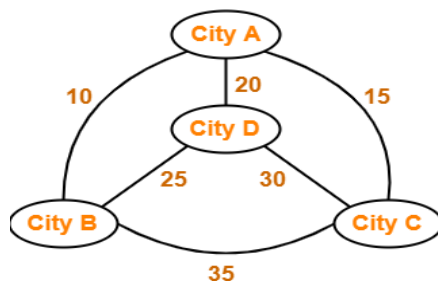
- Starting at vertex 'a' and resolving ties by the vertex alphabetical order, traverse the graph by depth-first search and construct the corresponding depth-first search tree. Give the order in which the vertices were reached for the first time (pushed onto the traversal stack) and the order in which the vertices became dead ends (popped off the stack).

17. **CO2** i) Apply heap sort using array representation for the list S,O,R,T,I,N,G in alphabetical order. (8)

CO1 ii) Construct a B-Tree of Order 3 by inserting numbers from 1 to 10 and find the B-Tree after deleting the numbers 5 & 6. (7)

18. **CO3** i) Analyze the NP- Clique problem. (5)

CO3 ii) The following graph shows a set of cities and distance between every pair of cities If salesman starting city is A, Test the minimum cost of the tour. (10)



19. **CO1** Build a C program for constructing an AVL tree with all the insertion and deletion operations for a given list of n distinct integers. (15)

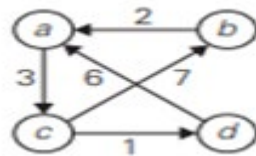
20. **CO3** Analyze the following Approximation algorithms problems. (15)

- Nearest –neighbor algorithm with an example
- Multifragment heuristic algorithm with an example.
- Minimum Spanning Tree algorithm

21. **CO2** i) Solve the following instance of the Knapsack problem by the branch-and-bound algorithm. Knapsack Capacity is 10. (8)

ITEM	WEIGHT	VALUE
1	4	Rs. 40
2	7	Rs. 42
3	5	Rs. 25
4	3	Rs. 12

- CO2** ii) Solve the Floyd's algorithm and find distance matrix for the following graph. (7)



22. **CO2** Construct Optimal BST for the following set of vertices along with its probability. (15)

A	B	C	D
0.2	0.1	0.3	0.4
